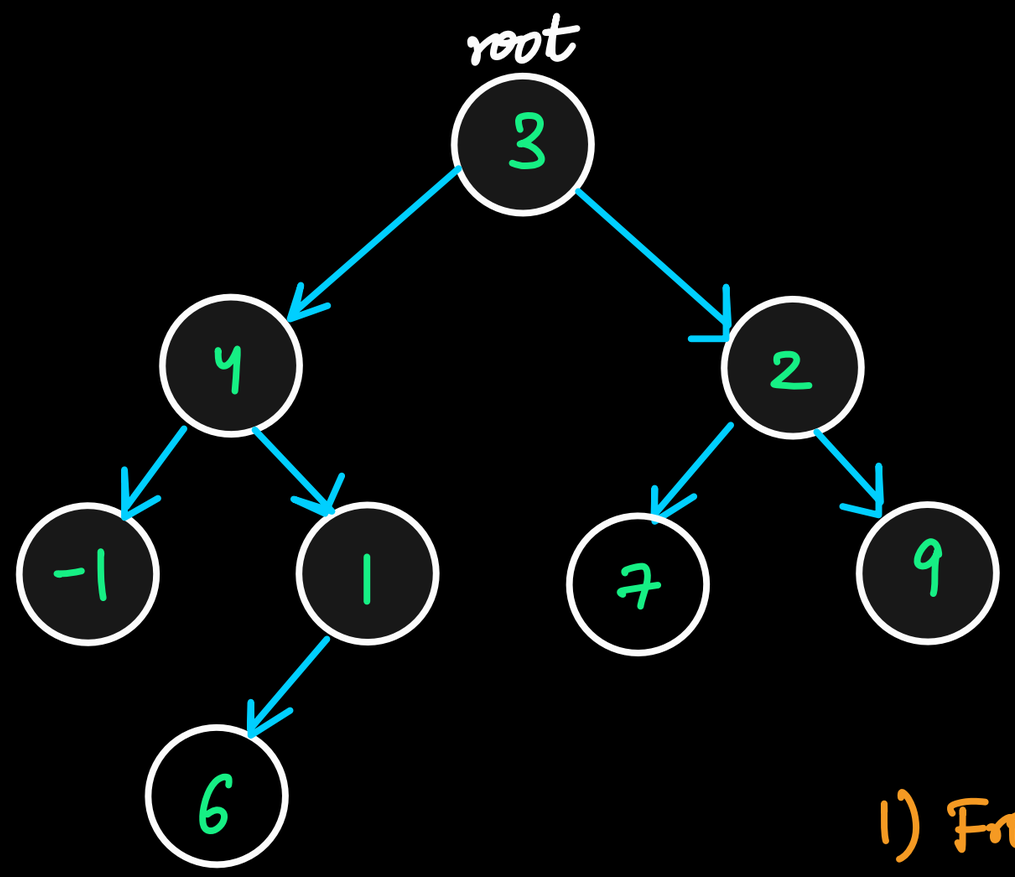
Level Order Traversal (BFS) (Breadth first search) (Queue)

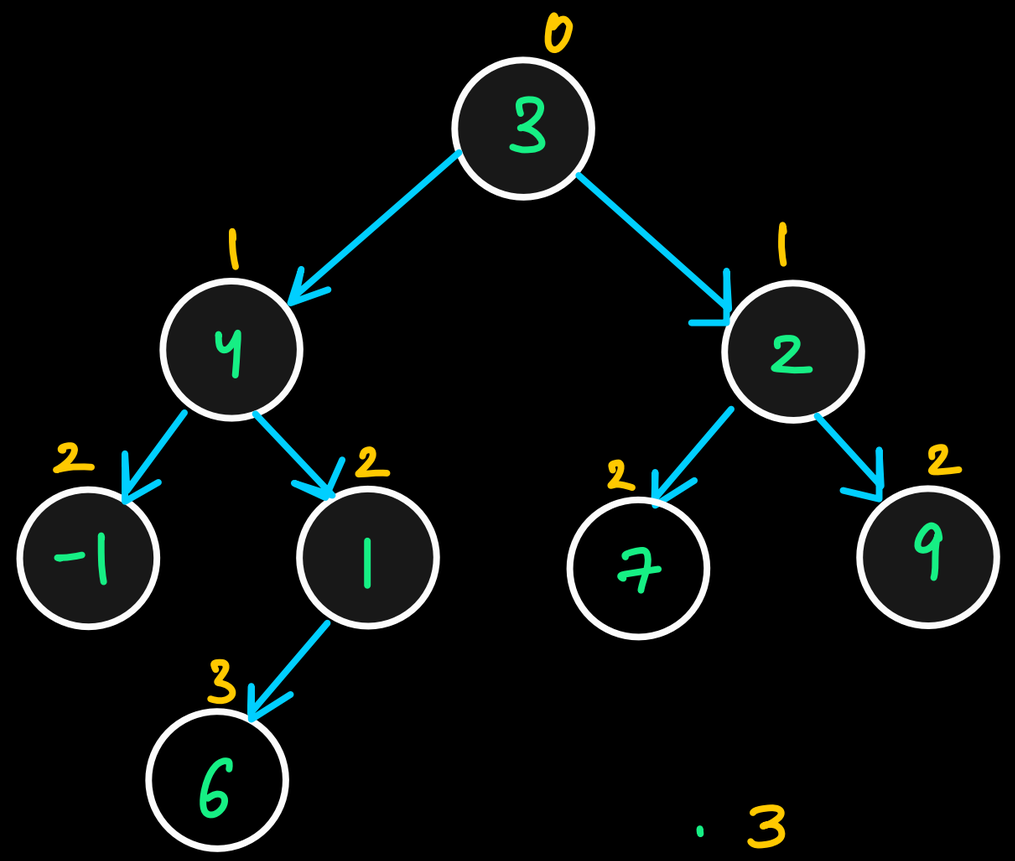


3 4 2 -1 1 7 9 6

1) Front se bahar nikalke print

2) Uske left & right child ko queue me daalo

Level Order Traversal



CurrLevel = ~~1~~ ~~2~~ ~~3~~ 4

- 3
- 4 2
- -1 1 7 9
- 6

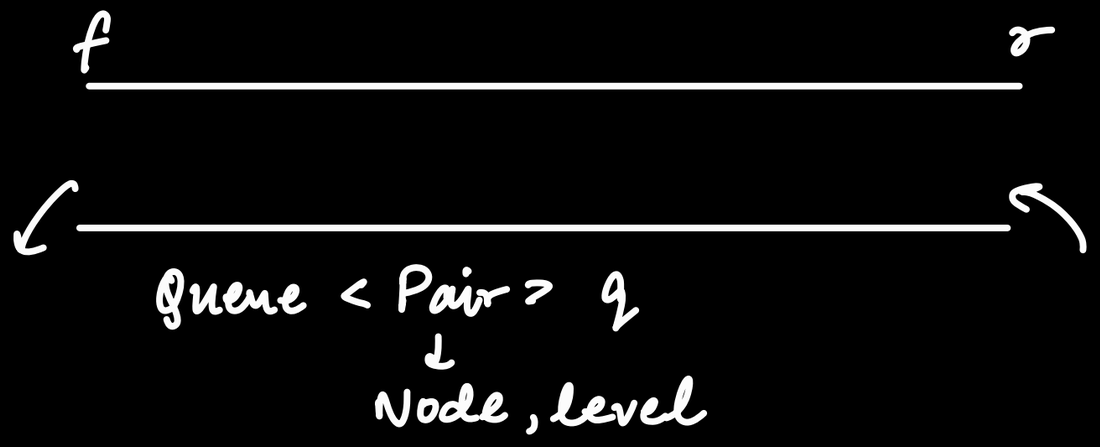
Output

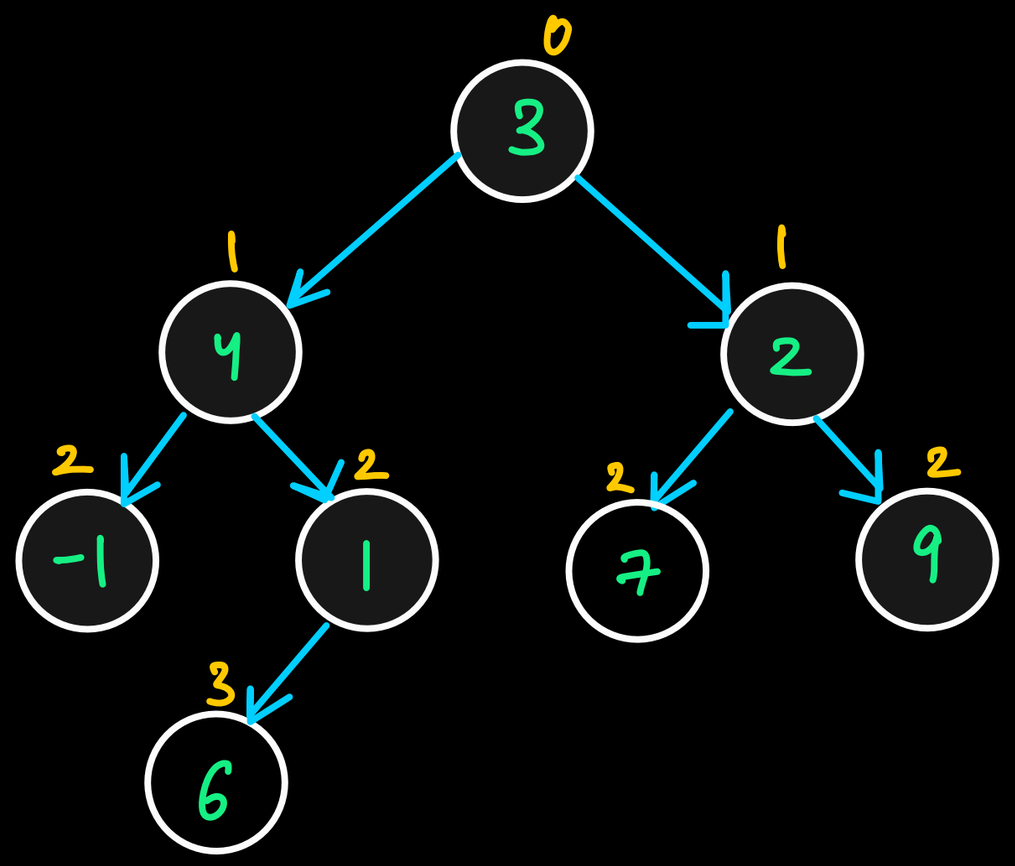
3

4 2

-1 1 7 9

6





3 4 -1 1 6 2 7 9

```
dfs(root, level){
    base case
    if(level == K) cout(root.val);
    dfs(root.left, level+1)
    dfs(root.right, level+1)
}
```

3

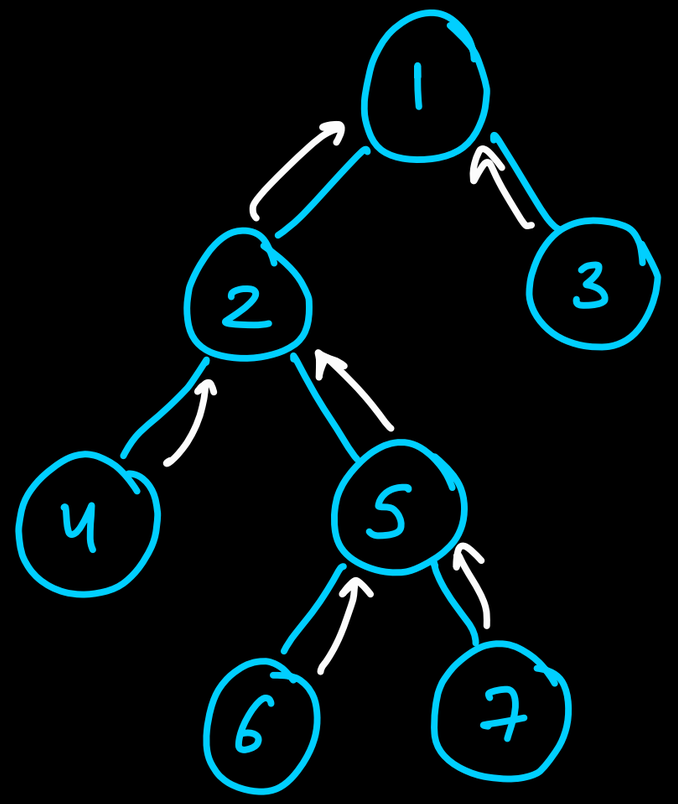
H.W. Zig-Zag Tree Traversal

H.W. Full Binary Tree

H.W. Perfect Binary tree

GFG Practice

Ques: Binary Tree Paths



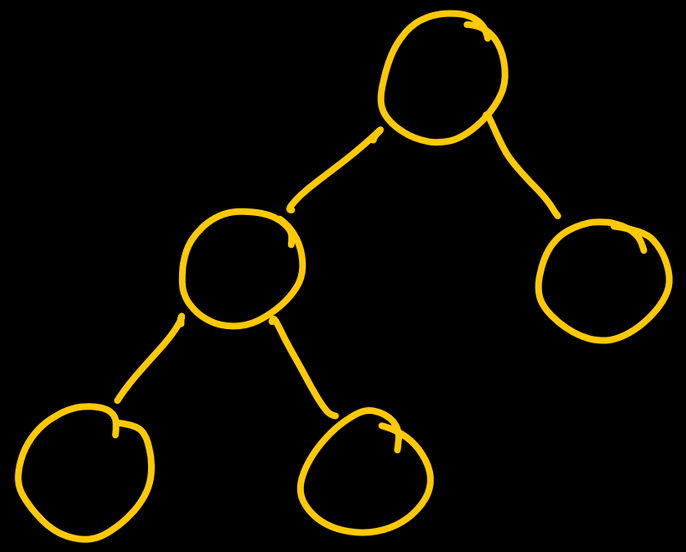
[ ]

```
fun(root, ArrayList<>){
    add
    fun(root.left)
    fun(root.right)
    remove
```

ans = { {1,2,4} , {1,2,5,6} , {1,2,5,7} , {1,3} }

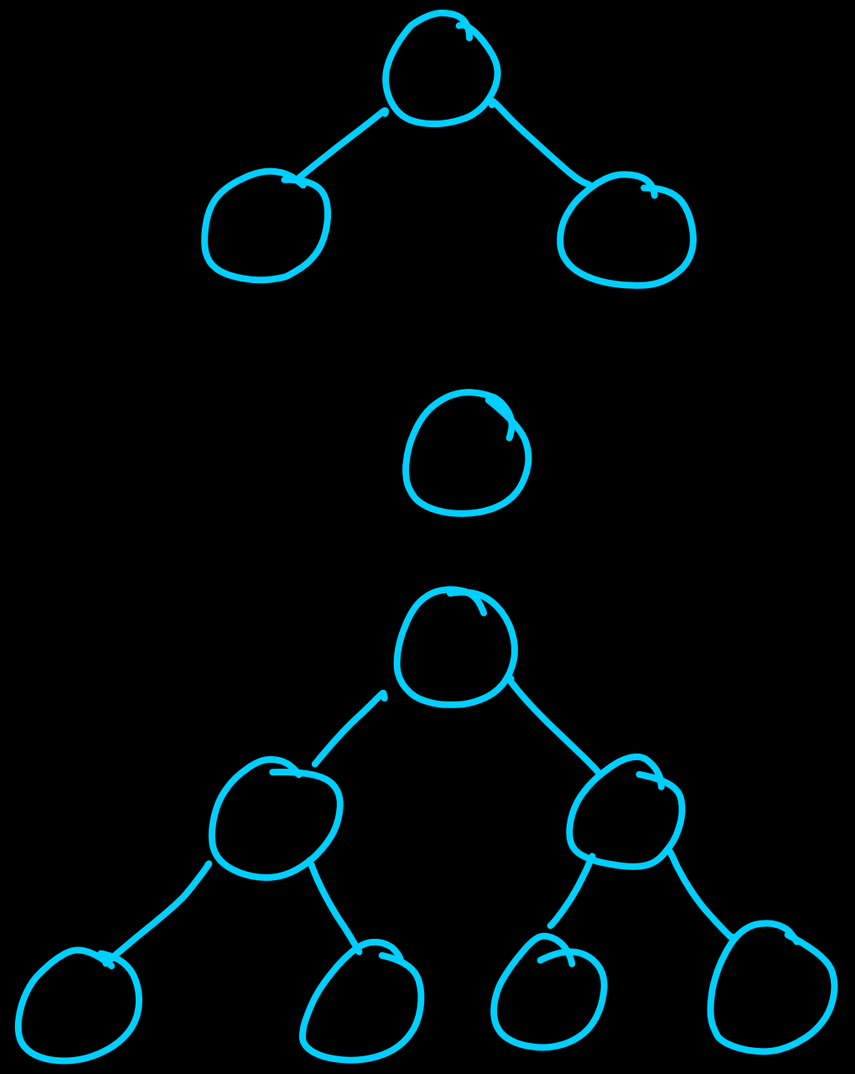
Types of Binary Trees

1. Full Binary Tree (0 or 2)
2. Perfect Binary Tree
3. Complete Binary Tree
4. Balanced Binary Tree
5. Degenerate & Skew Binary Trees
6. Binary Search Trees



Types of Binary Trees

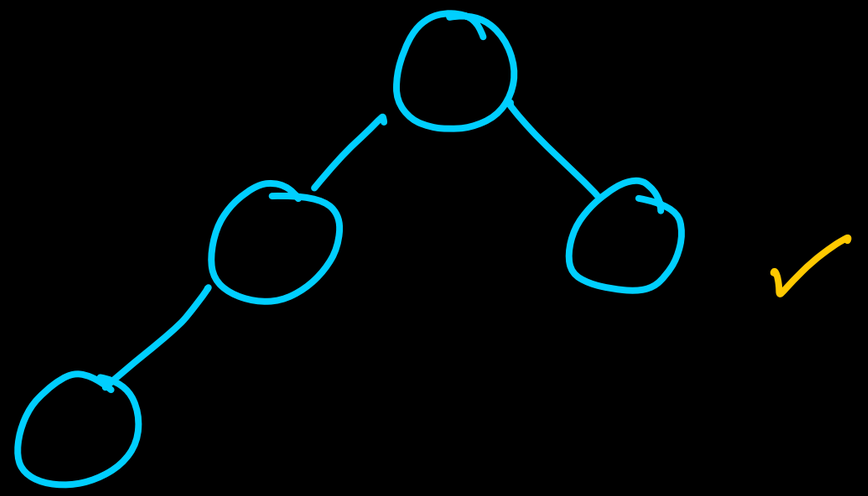
1. Full Binary Tree (0 or 2)
2. Perfect Binary Tree
3. Complete Binary Tree
4. Balanced Binary Tree
5. Degenerate & Skew Binary Trees
6. Binary Search Trees

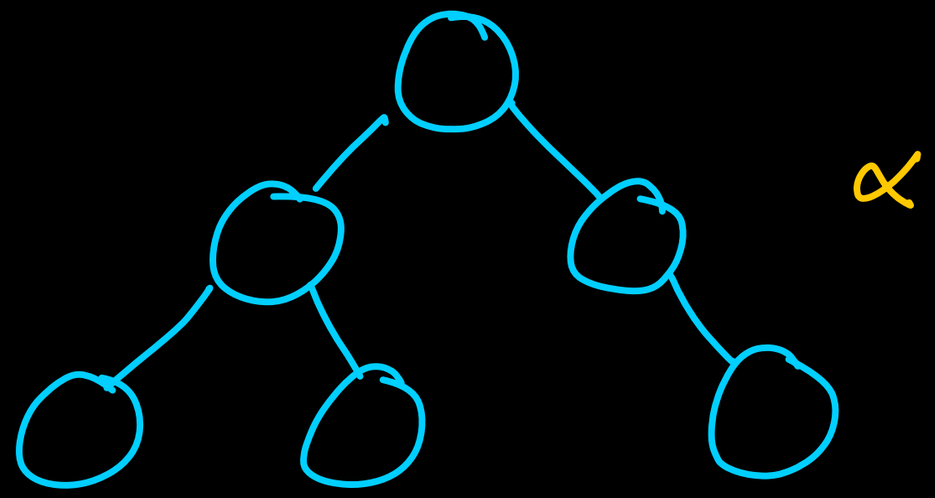


Types of Binary Trees

1. Full Binary Tree (0 or 2)
2. Perfect Binary Tree
3. Complete Binary Tree (CBT)
4. Balanced Binary Tree
5. Degenerate & Skew Binary Trees
6. Binary Search Trees

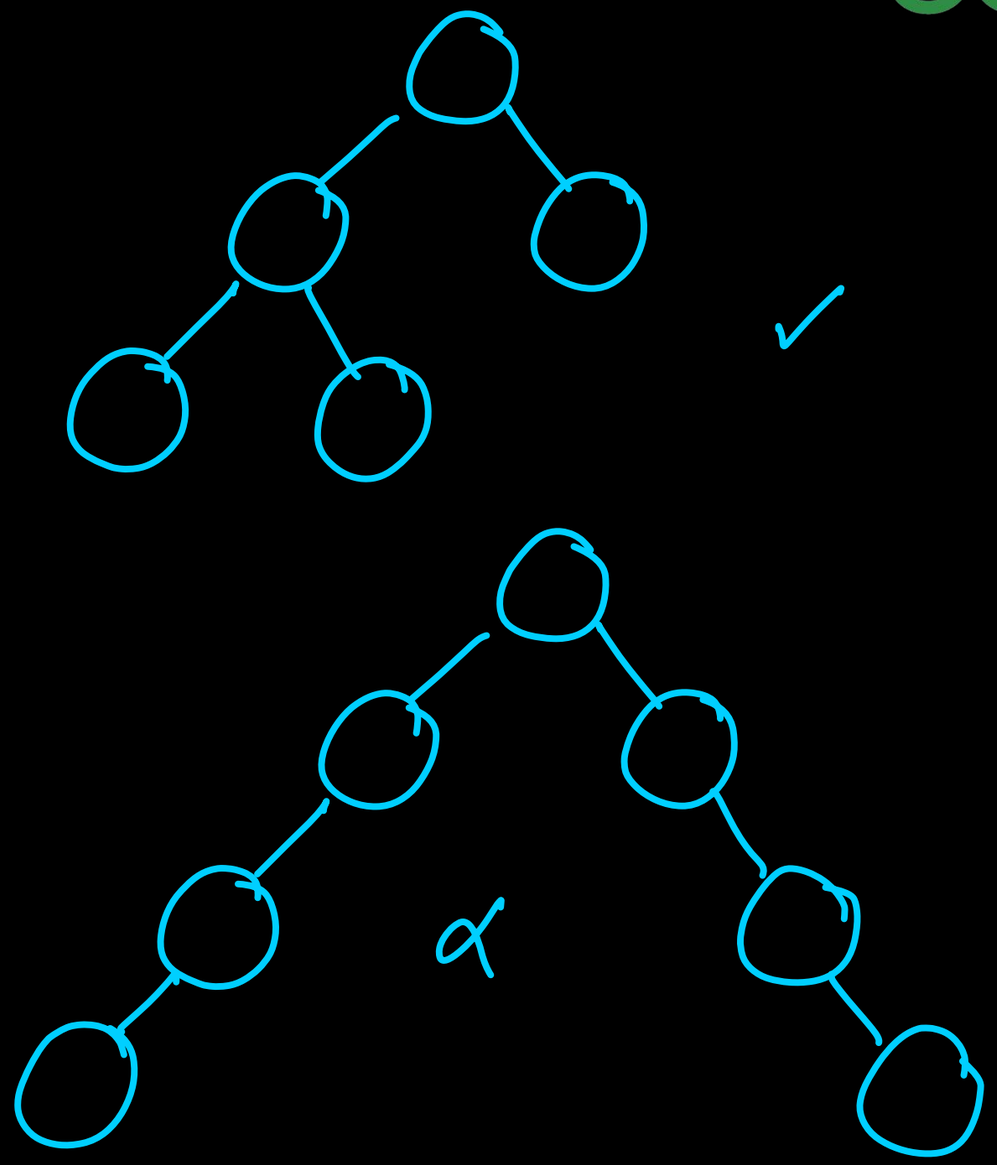
Last level ko chhode baaki
saari full/perfect
left to right filling





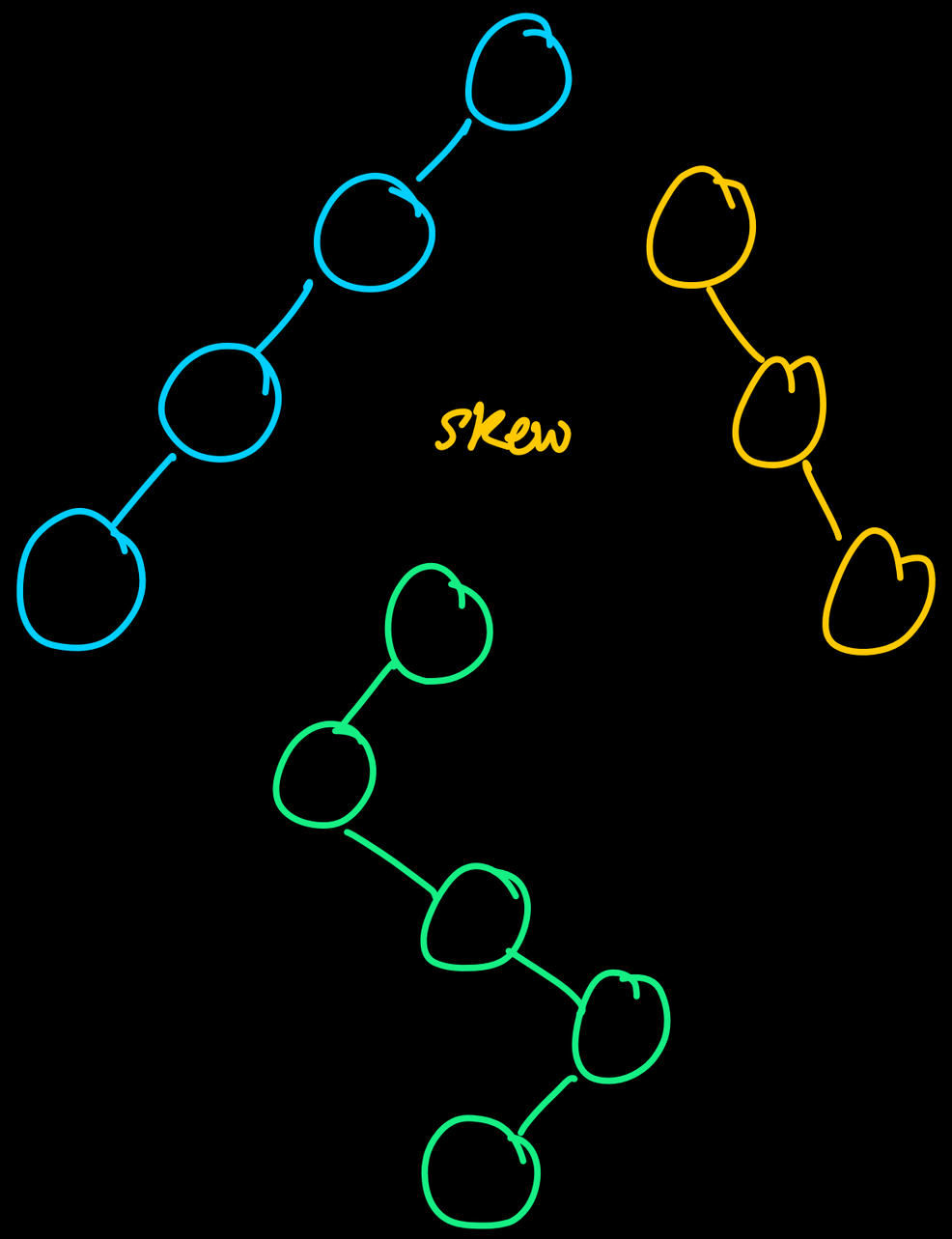
Types of Binary Trees

1. Full Binary Tree (0 or 2)
2. Perfect Binary Tree
3. Complete Binary Tree
4. Balanced Binary Tree
5. Degenerate & Skew Binary Trees
6. Binary Search Trees



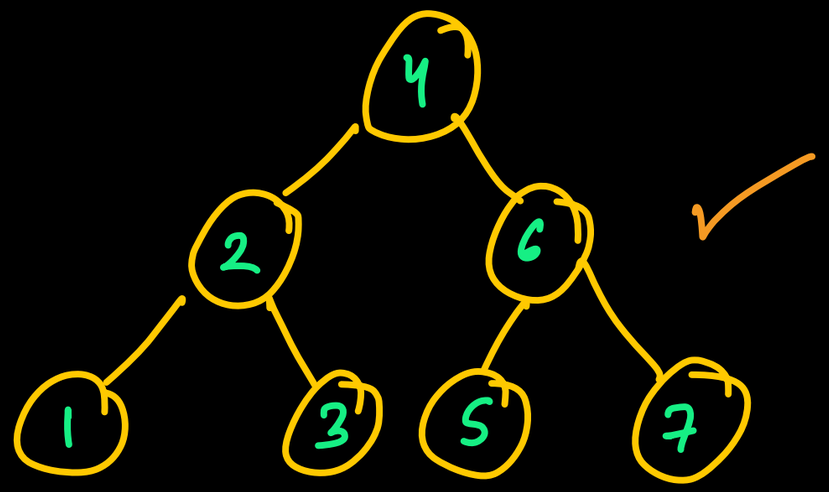
Types of Binary Trees

1. Full Binary Tree (0 or 2)
2. Perfect Binary Tree
3. Complete Binary Tree
4. Balanced Binary Tree
5. Degenerate & Skew Binary Trees
6. Binary Search Trees



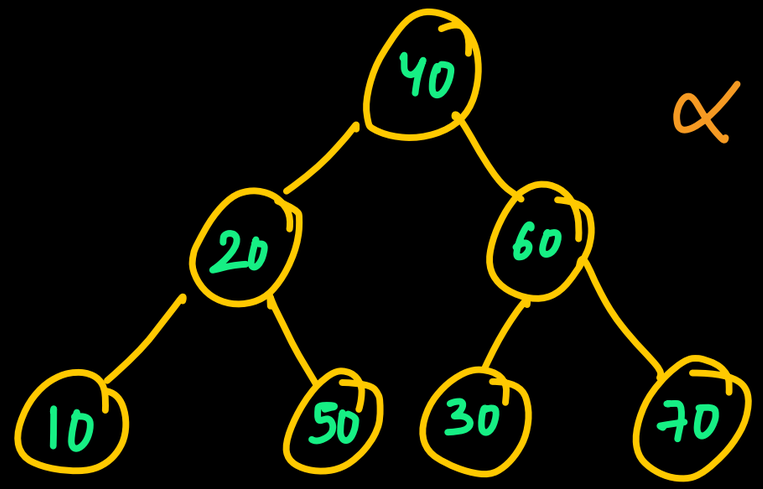
Types of Binary Trees

1. Full Binary Tree (0 or 2)
2. Perfect Binary Tree
3. Complete Binary Tree
4. Balanced Binary Tree
5. Degenerate & Skew Binary Trees
6. Binary Search Trees



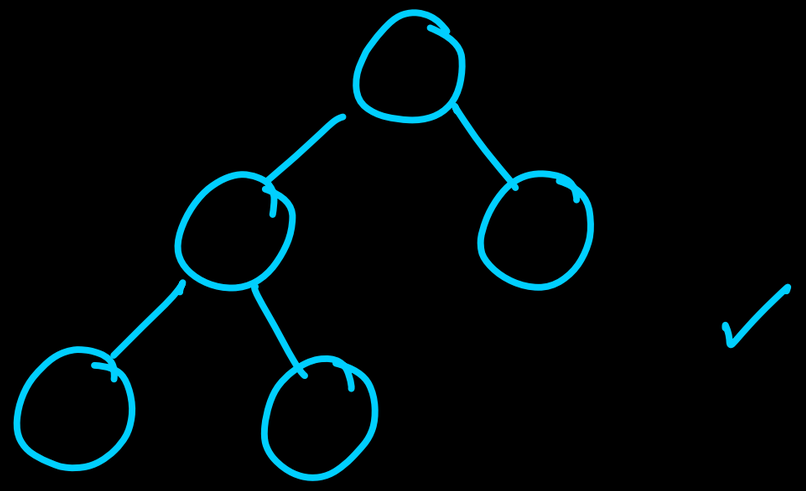
root.left.val < root.val < root.right.val ✗

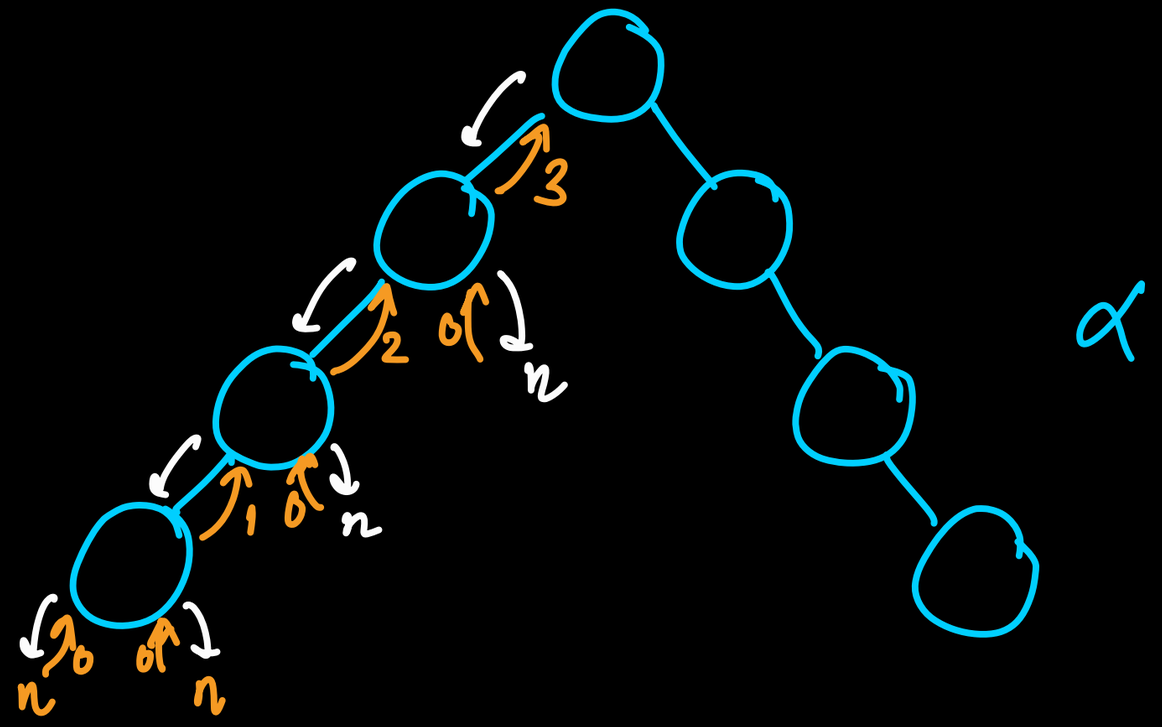
all val. in LST < root.val < All values in RST ✓



Ques: Balanced Binary Tree

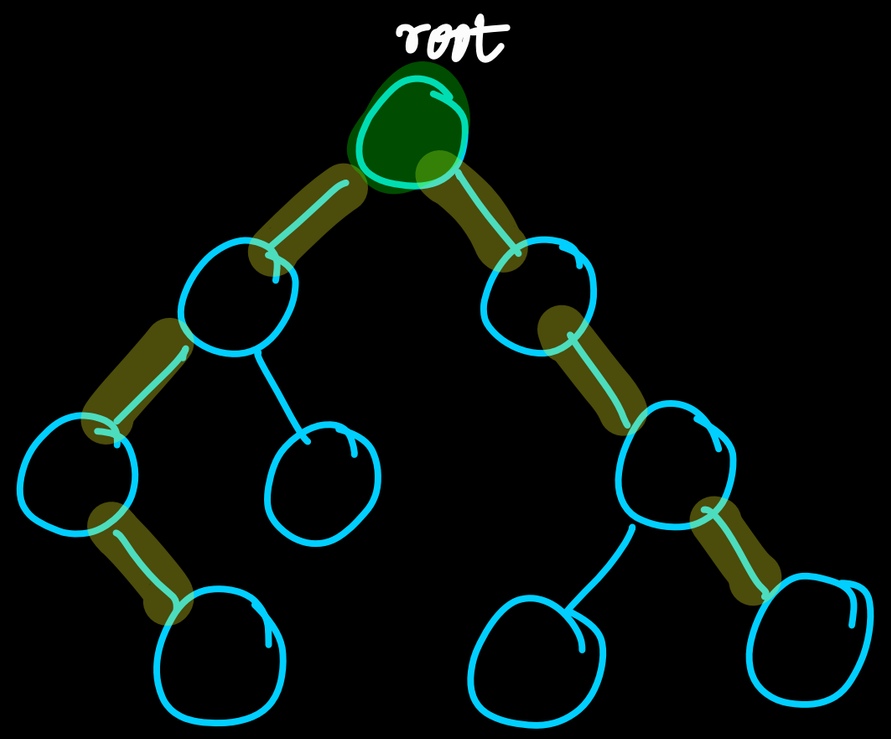
flag = false





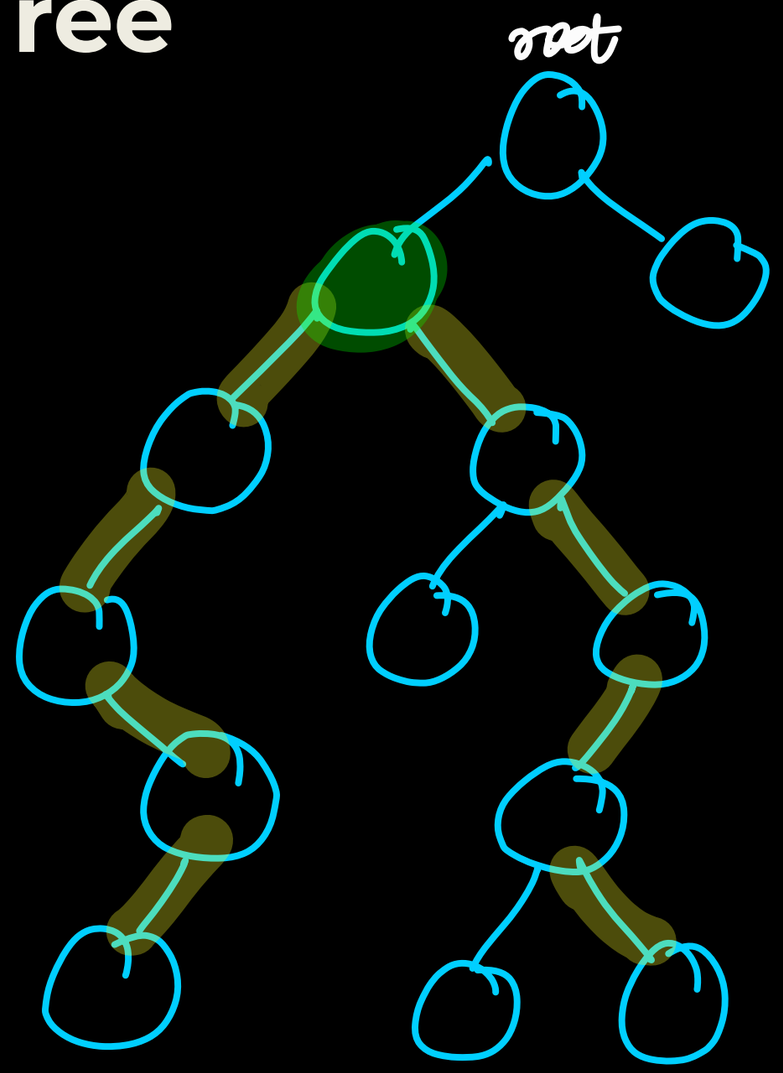
levels (LST) == levels (RST) & this should be valid for every node

Ques: Diameter of a Binary Tree



diameter = 6

path = left levels + right levels



diameter = 8